Arctic Sea Ice Keeps Shrinking and Getting Thinner

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Every year, the frozen seawater covering the Arctic Ocean grows during the long, dark winter and shrinks during the warmer months. It usually reaches its largest size in March and its smallest in September. **Satellites** have tracked this cycle since late 1978, giving scientists nearly 50 years of measurements.

Those measurements show the ice is shrinking at both ends of the cycle. In 2026, Arctic sea ice reached its smallest size on September 12, covering about 4.6 million square kilometers, or 1.78 million square miles. According to NASA and the National Snow and Ice Data Center, that tied 2008, 2010 and 2025 for the 10th-lowest level in the satellite record. The smallest ever was in September 2012. In winter, the March 2026 peak tied March 2025 for the lowest winter **maximum** on record.

Every summer from 2007 through 2026 ranks among the 20 lowest in the record. But over the past decade, the summer low has stayed fairly steady instead of dropping quickly. NASA researchers say more clouds over the Arctic in recent years blocked some sunlight and helped limit melting. Still, summer ice remains far below its **historical average**.

The ice itself is also changing. In the past, much of the Arctic held thick "**multiyear ice**," which survives at least one summer. Now more of the Arctic is covered by thinner ice that forms in a single winter and melts more easily.

Natural weather patterns affect the ice from year to year, but scientists say they cannot fully explain the long decline. Rising global temperatures, together with natural changes, have contributed to the loss. Scientists **project** the Arctic Ocean could become nearly ice-free in summer at least once before 2050.

Satellites with special **microwave** sensors let scientists see the ice even through clouds.

Word Treasure

satellites -- Machines that orbit Earth and collect information, such as pictures and measurements.

maximum -- The largest amount something reaches.

historical average -- The typical amount measured over many past years.

multiyear ice -- Sea ice that has lasted through at least one summer without melting, so it grows thicker.

project -- To estimate what will happen in the future using data and models.

microwave -- A type of invisible energy wave that special sensors can detect, even through clouds.

Background you need

Sea ice is frozen ocean water that floats on the surface, unlike icebergs, which break off glaciers on land.

Bright white ice reflects much of the sunlight that hits it, while dark open ocean absorbs more heat.

When sea ice melts, more dark water is exposed, which can warm the region further.

Scientists compare each year with long satellite records to see whether changes are part of a lasting trend.

Break it down

LEAD: Nearly 50 years of satellite records show Arctic sea ice shrinking at both ends of its yearly cycle

KEY EVENT: The 2026 summer minimum tied for 10th-lowest; the March 2026 peak tied for lowest winter maximum

+- WHO: NASA and the National Snow and Ice Data Center

+- WHEN: Minimum reached September 12, 2026

EVIDENCE: Every summer from 2007 to 2026 ranks among the 20 lowest; thick multiyear ice is being replaced by thinner ice

NUANCE: Summer losses slowed over the past decade, partly because of more clouds

OPEN QUESTION: When might the Arctic have its first nearly ice-free summer?

Why it matters

Arctic sea ice is a key part of Earth's climate system, and its long decline shows how rising temperatures are changing a region that has had ice cover for thousands of years.

Test what you remember

1. When did Arctic sea ice reach its smallest size in 2026?

- (A) March 12
- (B) December 12
- (C) June 12
- (D) September 12

2. How large was the sea ice at its 2026 minimum?

- (A) About 20 million square kilometers
- (B) About 1.78 million square kilometers
- (C) About 4.6 million square kilometers
- (D) About 10 million square kilometers

3. In which year was the smallest summer ice ever observed?

- (A) 2026
- (B) 2025
- (C) 2008
- (D) 2012

4. What do NASA researchers say helped limit melting in recent years?

- (A) Stronger winds
- (B) Colder oceans
- (C) More clouds blocking sunlight
- (D) Fewer ships

5. How is the Arctic's ice changing besides covering less area?

- (A) It is turning darker
- (B) It is becoming younger and thinner
- (C) It is becoming older and thicker
- (D) It is moving south

6. What do scientists project could happen before 2050?

- (A) Satellites stop working
- (B) A nearly ice-free summer at least once
- (C) The Arctic gains more ice
- (D) Winter ice disappears completely

Answer key (try the quiz first!): 1: D 2: C 3: D 4: C 5: B 6: B

Questions to think about

1. NASA researchers: Say increased cloud cover is one factor behind the recent slowdown in summer ice loss.

2. Scientists: Say natural patterns cannot fully explain the decline and that rising temperatures have contributed.

MY THOUGHTS (at least 20 words):